Date
Course

Practice Test II (Version 2)

I) Write your name, the date, and the course title (i.e. Precalculus).
II) Show all work.
III) Draw a rectangle around your final answer for each problem.
IV) You may use a calculator except for graphing.

① Find the sum of all of the terms in the geometric sequence.

100, 12.5, 1.5625, ...

② Find the sum. You may write your answer in exponential form.

$$\sum_{n=2}^{41} -0.75^{n+1}$$

③ i) Find the sum.

$$\sum_{n=0}^{3} 2(n+3)-4$$

ii) Write an equation of the n^{th} term of the sequence below in recursive form. Also, show the value of the first term.

$-\frac{1}{10}, 0, \frac{1}{10}, \frac{1}{5}, \ldots$

④ A man walks into a donut shop and chooses a donut at random. The probability that the donut will have some sort of chocolate is 23%. The probability that the donut will have sprinkles is 7%. The probability that the donut will have both chocolate and sprinkles is 3.3%. What is the probability that the donut will have either chocolate or sprinkles?

5 In a garage there are 6 wrenches and 8 screwdrivers. 4 of these tools are chosen at random. What is the probability that at least one will be a wrench?

6 Find the third term in the expansion of $(x-3y)^5$.

7 i) Find the 97^{th} term in the arithmetic sequence below.

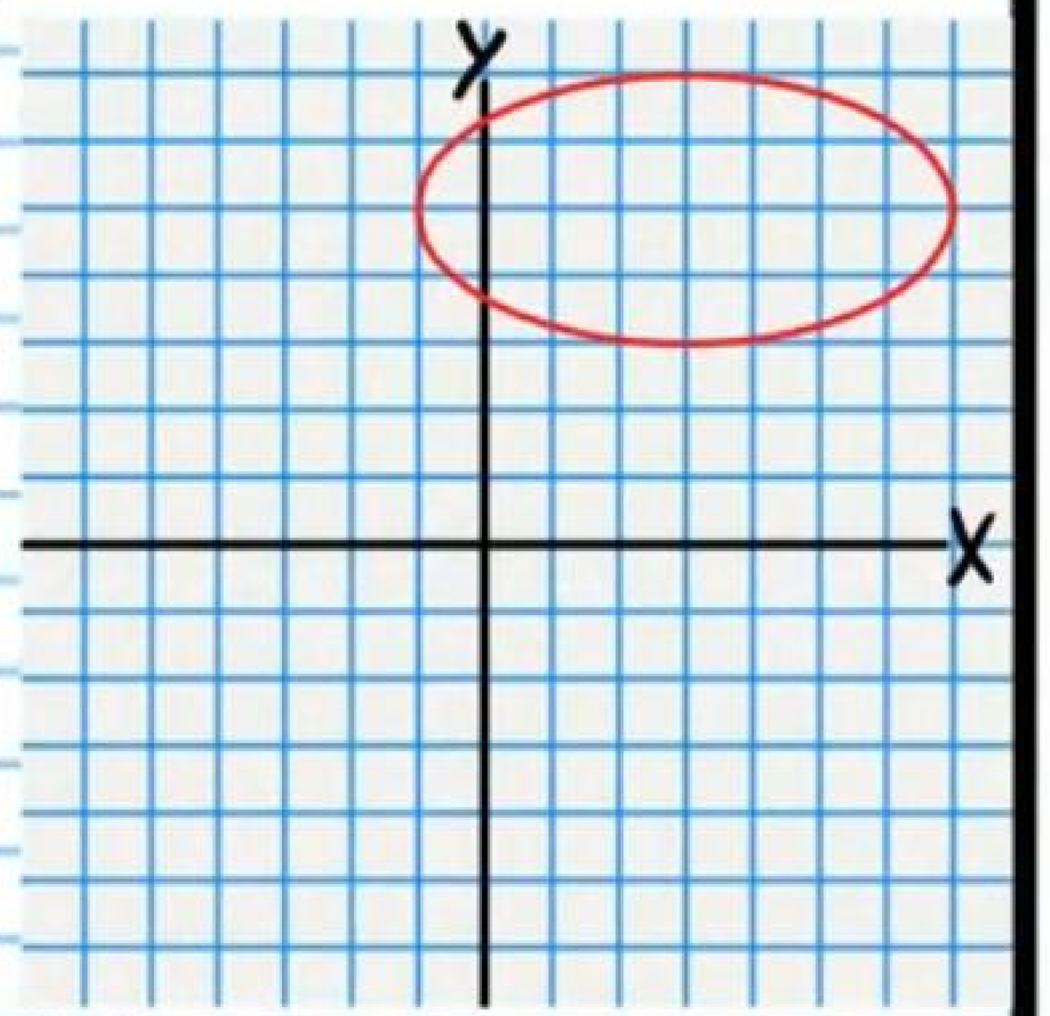
$-40, -33, -26, \ldots$

ii) Find the 23^{rd} term in the geometric sequence below. You may leave your answer in exponential form.

$-1, 6, -36, \ldots$

⑧ A restaurant offers Coca-Cola, water, and iced tea. It sells hot dogs and burgers for the main course and sour cream chips, orange puffs, corn chips, apples, and oranges for a side. If these are the only items that the restaurant sells, and a customer will choose one drink, one main course, and one side, how many outcome variations are possible for the customer's meal?

⑨ Write an equation of the graph below.

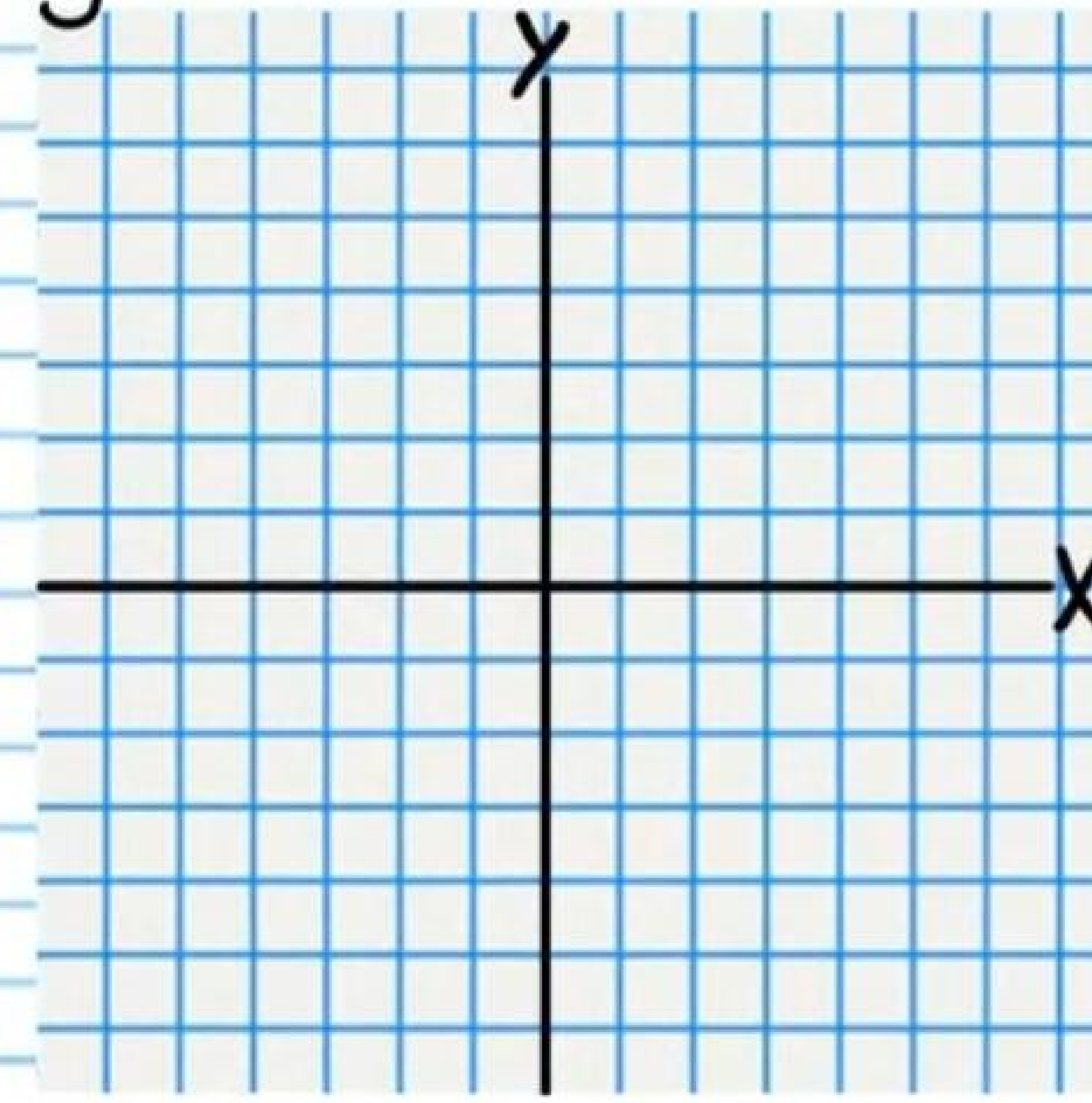


⑩ Use the Binomial Theorem to expand the following expression.

$(a+3b)^4$

⑪ Graph the following equation. Write the coordinates of the center and the length of the radius.

$1=(x-4)^2+(y-5)^2$



⑫ Billy has 10 distinct pairs of pants, 8 distinct shirts, 8 distinct pairs of shoes, and 15 distinct hats. He is packing for his vacation, but he will only take 5 pairs of pants, 6 shirts, 3 pairs of shoes, and 10 hats. How many outcome variations are there?

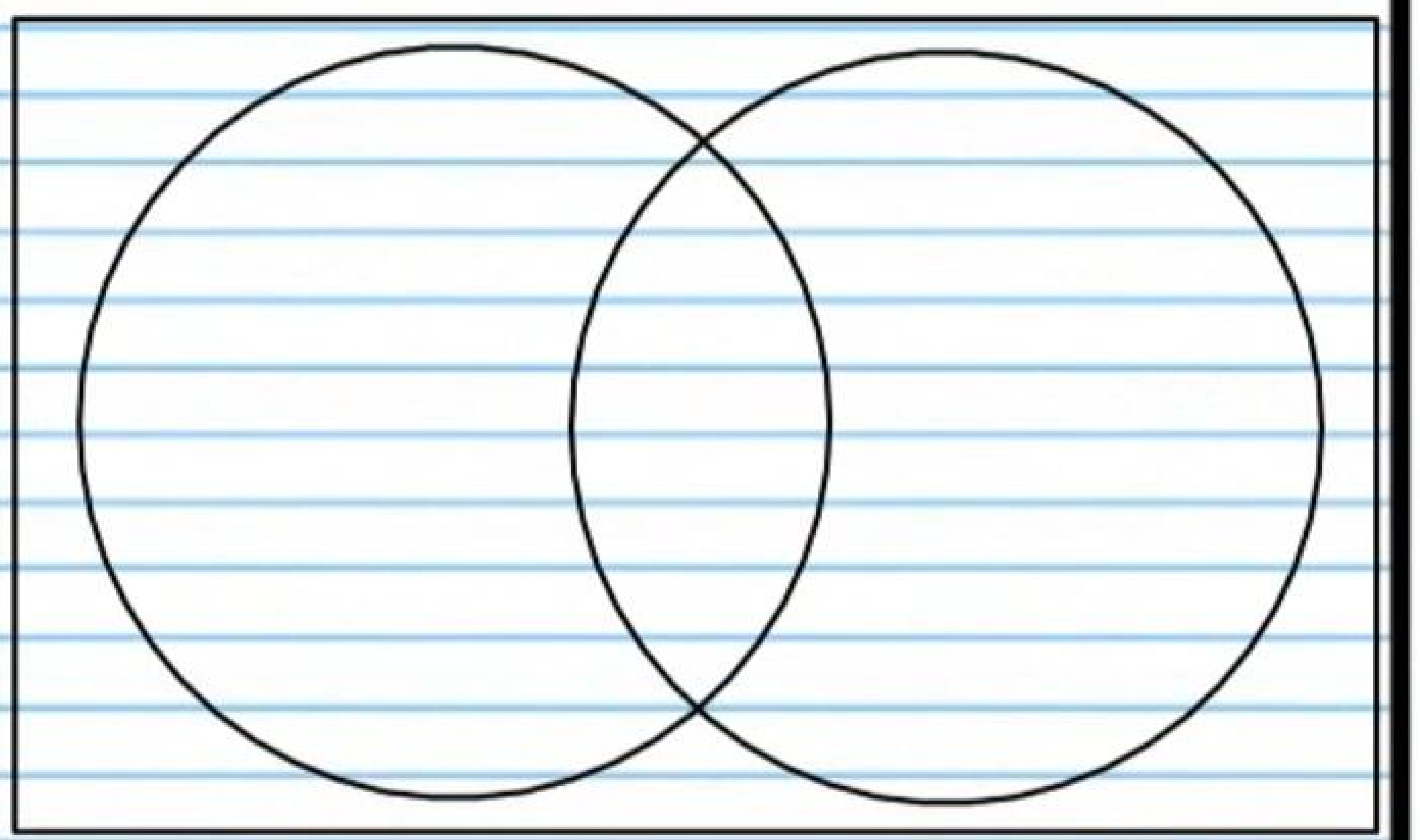
⑬ A family of 29 went on vacation in Hawaii. 13 people surfed. 22 people scuba dived. 8 people did both.

i) How many people surfed but did not scuba dive?

ii) How many people scuba dived but did not surf?

iii) How many people either surfed or scuba dived?

iv) How many people neither surfed nor scuba dived?



v) What is the probability that a randomly chosen person scuba dived?

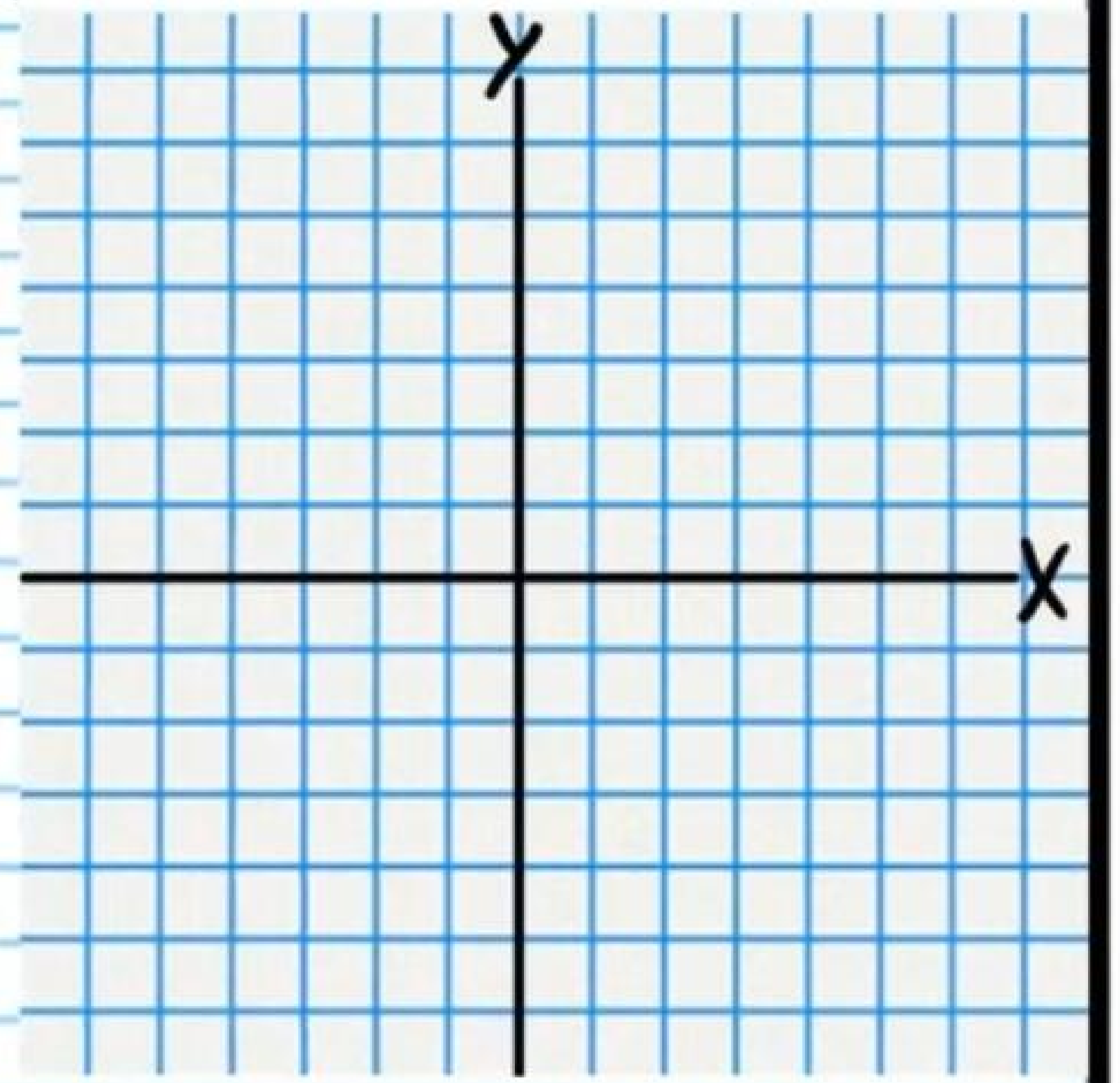
vi) What is the probability that a randomly chosen person neither surfed nor scuba dived?

vii) What is the probability that a randomly chosen person both surfed and scuba dived?

viii) What is the probability that a randomly chosen person either surfed or scuba dived?

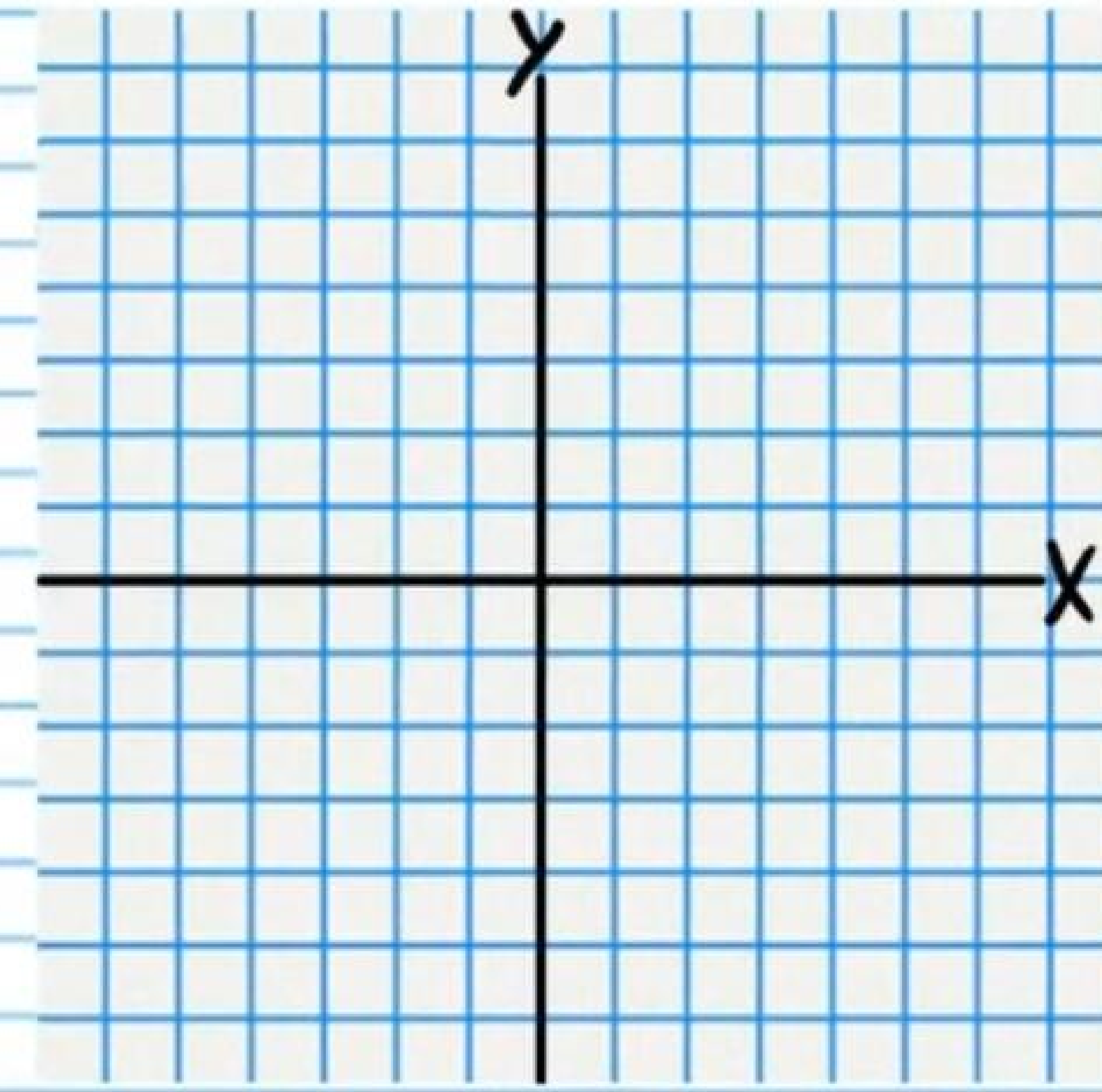
⑭ Graph the following equation. ☆ You must first convert to a recognizable form.

$4y^2-9x^2+40y+64=0$



⑮ Graph the following equation. ☆ You must first convert to a recognizable form.

$2y^2+16y-3x+23=0$



For problem #'s 16–17, determine if each sequence is arithmetic, geometric, or neither. Then find an expression for the n^{th} term. Assume that the first term is produced when $n=1$.

(16) i) $10, 1, -8, \ldots$

ii) $1, 8, 27, 64, \ldots$

(17) i) $\frac{3}{5}, \frac{2}{5}, \frac{1}{5}, \ldots$

ii) $0.3, 0.1, 0.0\overline{3}$

(18) i) Find the sum of the first 39 terms of the arithmetic sequence below.

$90, 75, 60, \ldots$

ii) 13 students try out for the school baseball team. There are nine distinct positions. How many possible outcome variations are there if only nine students can be chosen? Note: A team where John pitches is different from a team where Bill pitches.

(19) Let $A=\{-16,-15,-14,-13,-12\}$ and $B=\{-14,-13,-12,-11\}$.

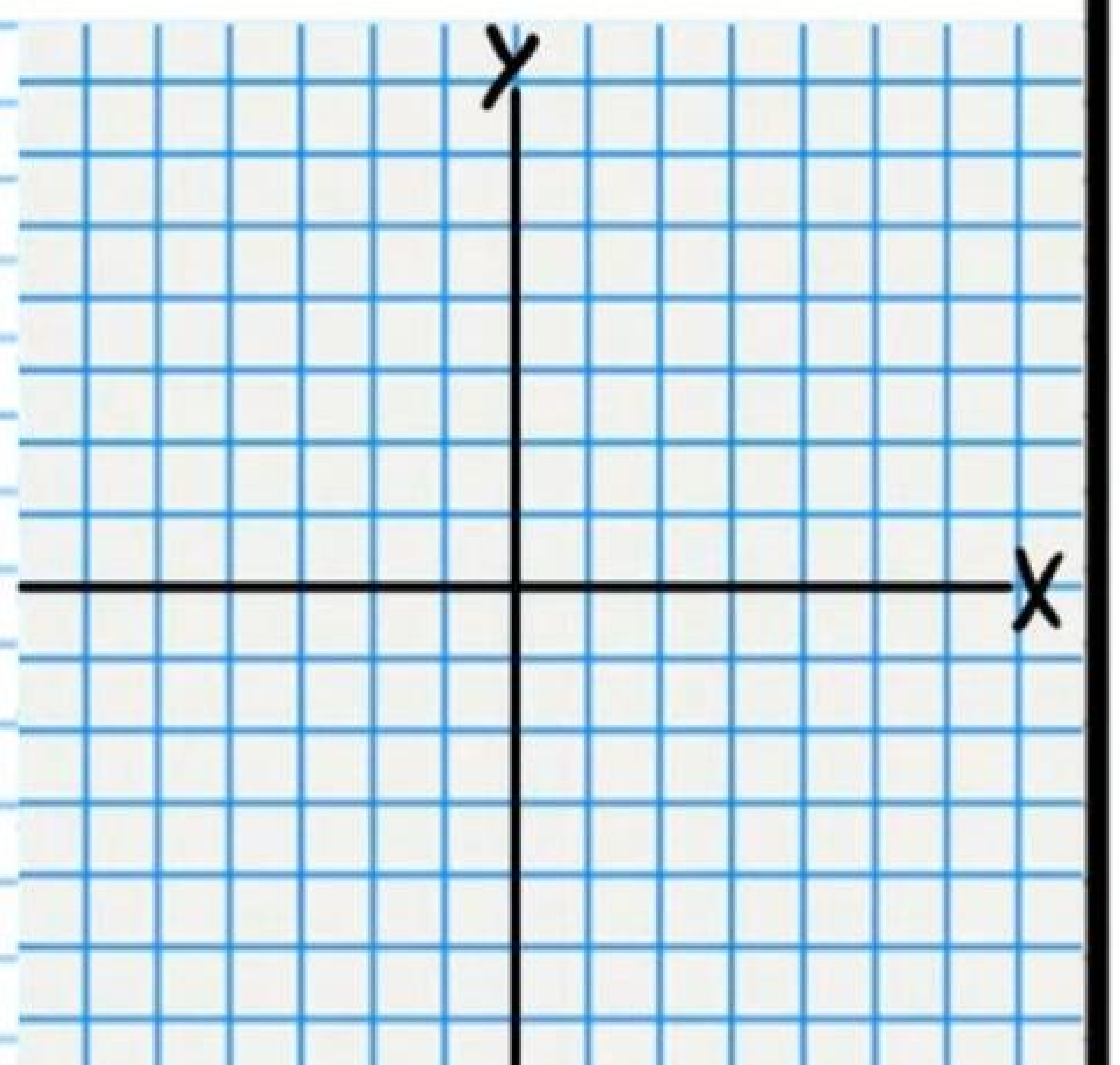
i) Find $A \cap B$

ii) Find $A \cup B$

(20) If the 6^{th} term of a geometric sequence is $-\frac{1}{5}$, and the 9^{th} term is 25, what is the 3^{rd} term?

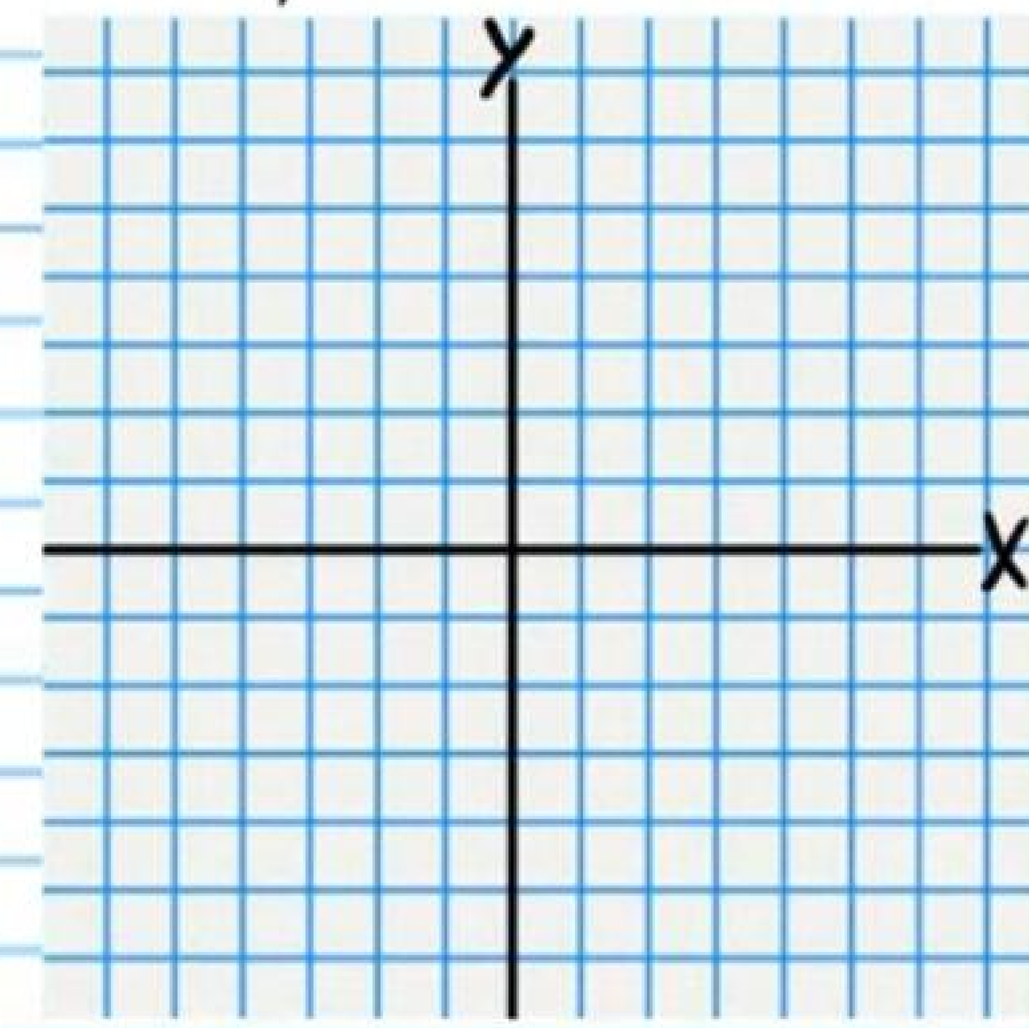
(21) Graph the following equation. Write the coordinates of the foci.

$$1=\frac{x^2}{25}+\frac{y^2}{36}$$

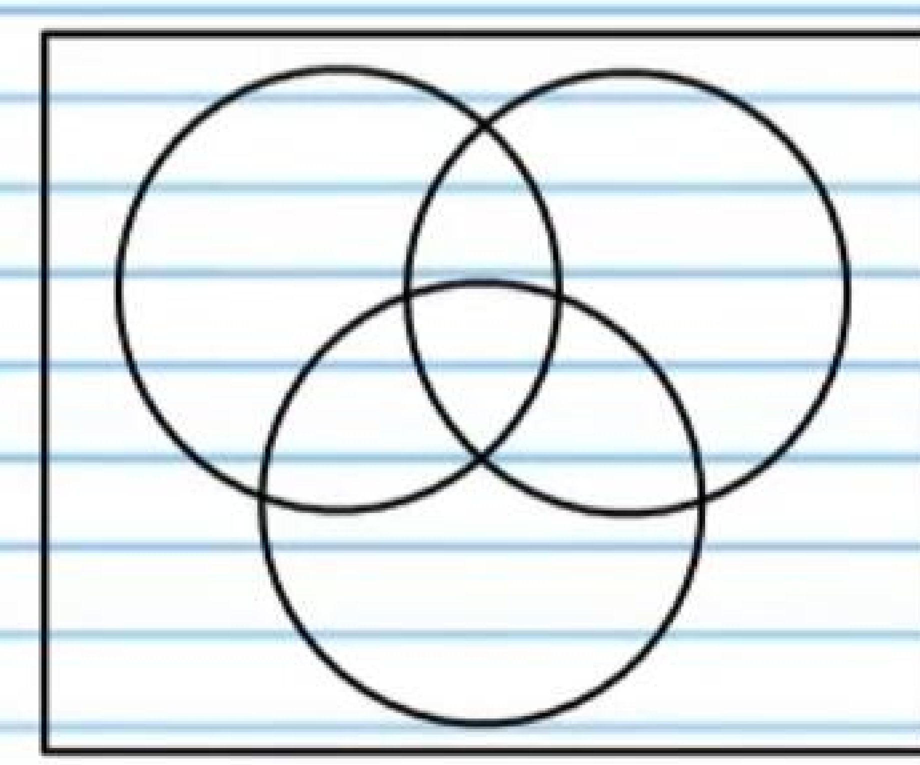


(22) Find an equation of the parabola with the following characteristics and graph the equation.

Focus at $(-2, 1\frac{7}{8})$ and directrix $y=2$.

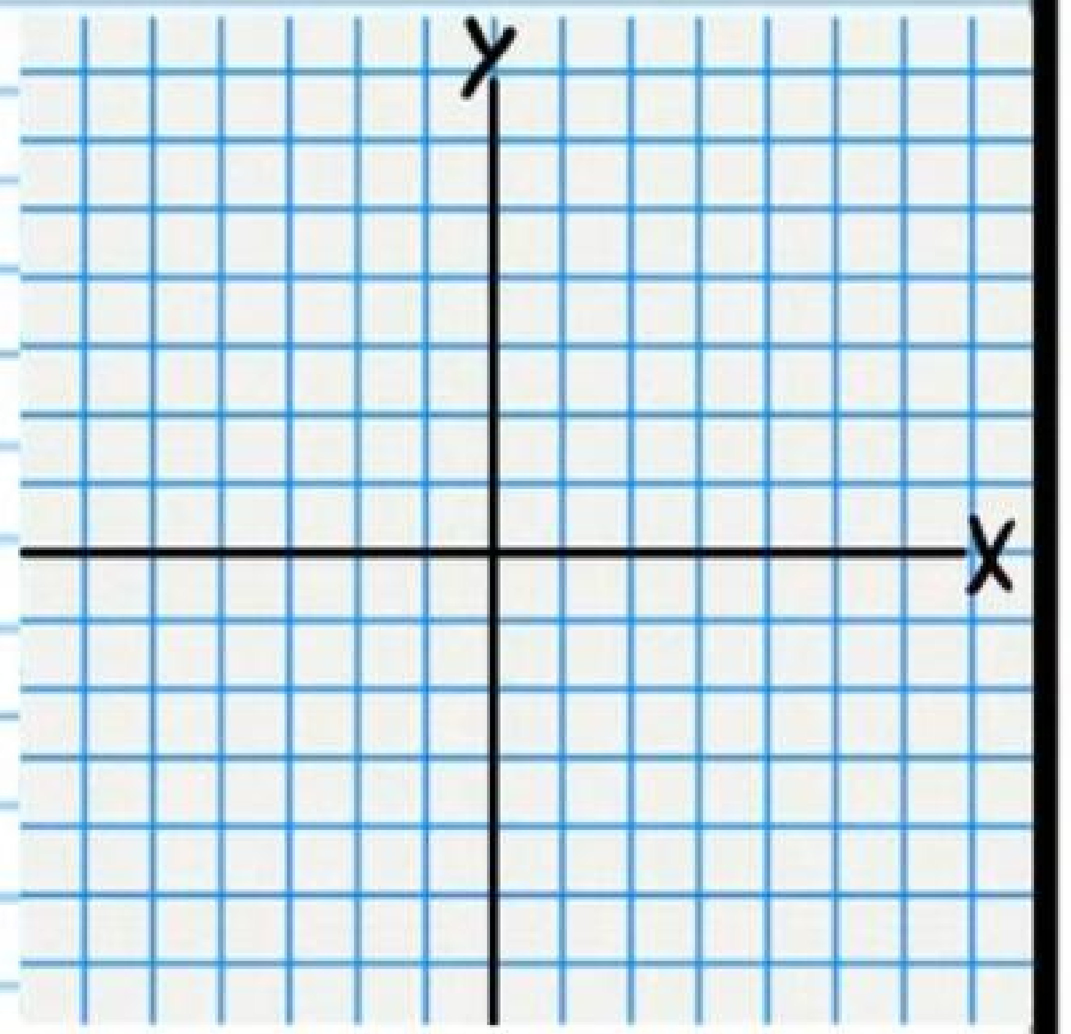


(23) A business sends fruit baskets for gift-givers. They sent 60 baskets this month. 32 of the baskets contained grapes. 28 of the baskets contained apples. 21 of the baskets contained apples and oranges. 16 of the baskets contained oranges and grapes. 14 of the baskets contain grapes and apples. 9 baskets contain all three types of fruit and 7 baskets had none of these types of fruit. How many baskets had oranges but no apples or grapes?



(24) Graph the following equation. Write the coordinates of the foci, draw the asymptotes as dotted lines, and write their equations.

$$\frac{(x+1)^2}{8} - \frac{(y-2)^2}{8} = 1$$



Asymptotes

(25) There are 11 blue marbles, 6 green marbles, and 7 red marbles in a bag. 4 marbles are chosen at random.

i) What is the probability that all 4 are green?

ii) What is the probability that 3 marbles will be red and 1 will be blue?

iii) What is the probability that 2 will be green and 2 will be red?

iv) What is the probability that 2 marbles will be green, 1 will be blue, and 1 will be red?

Practice Test II (Version 2) Answers

① $\boxed{\frac{800}{7}}$ or $\boxed{114\frac{2}{7}}$ ② $\boxed{-0.75^{3}\frac{1-(0.75)^{40}}{0.25}}$ or $\boxed{-\frac{27}{16}\left(1-\left(\frac{3}{4}\right)^{40}\right)}$

③ i) $\boxed{20}$ ii) $\boxed{\left\{a_1=-\frac{1}{10},\ a_n=a_{n-1}+\frac{1}{10}\right\}}$

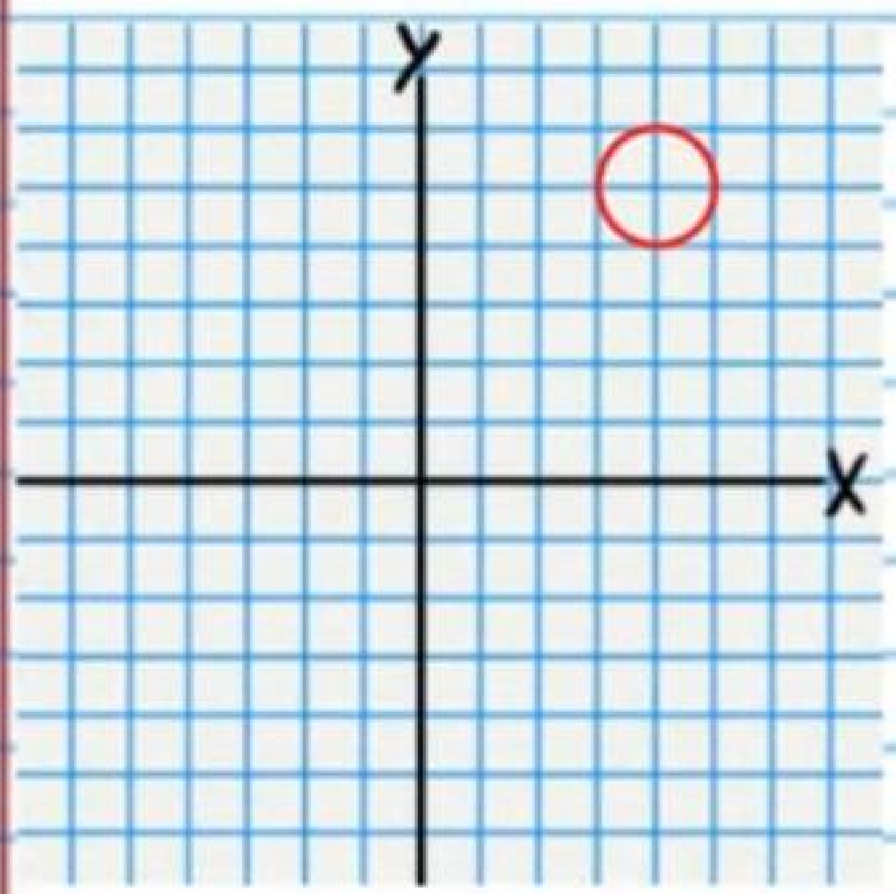
④ $\boxed{0.267}$ or $\boxed{26.7\%}$ or $\boxed{\frac{267}{1000}}$ ⑤ $\boxed{\frac{133}{143}}$ ⑥ $\boxed{90x^3y^2}$

⑦ i) $\boxed{632}$ ii) $\boxed{-(-6)^{22}}$ or $\boxed{-6^{22}}$

⑧ $\boxed{\text{30 outcome variations}}$ ⑨ $\boxed{\frac{(x-3)^2}{16}+\frac{(y-5)^2}{4}=1}$

⑩ $\boxed{a^4+12a^3b+54a^2b^2+108ab^3+81b^4}$

⑪ $\boxed{c(4,5),\ r=1}$ ⑫ $\boxed{\text{1,186,593,408 outcome variations}}$

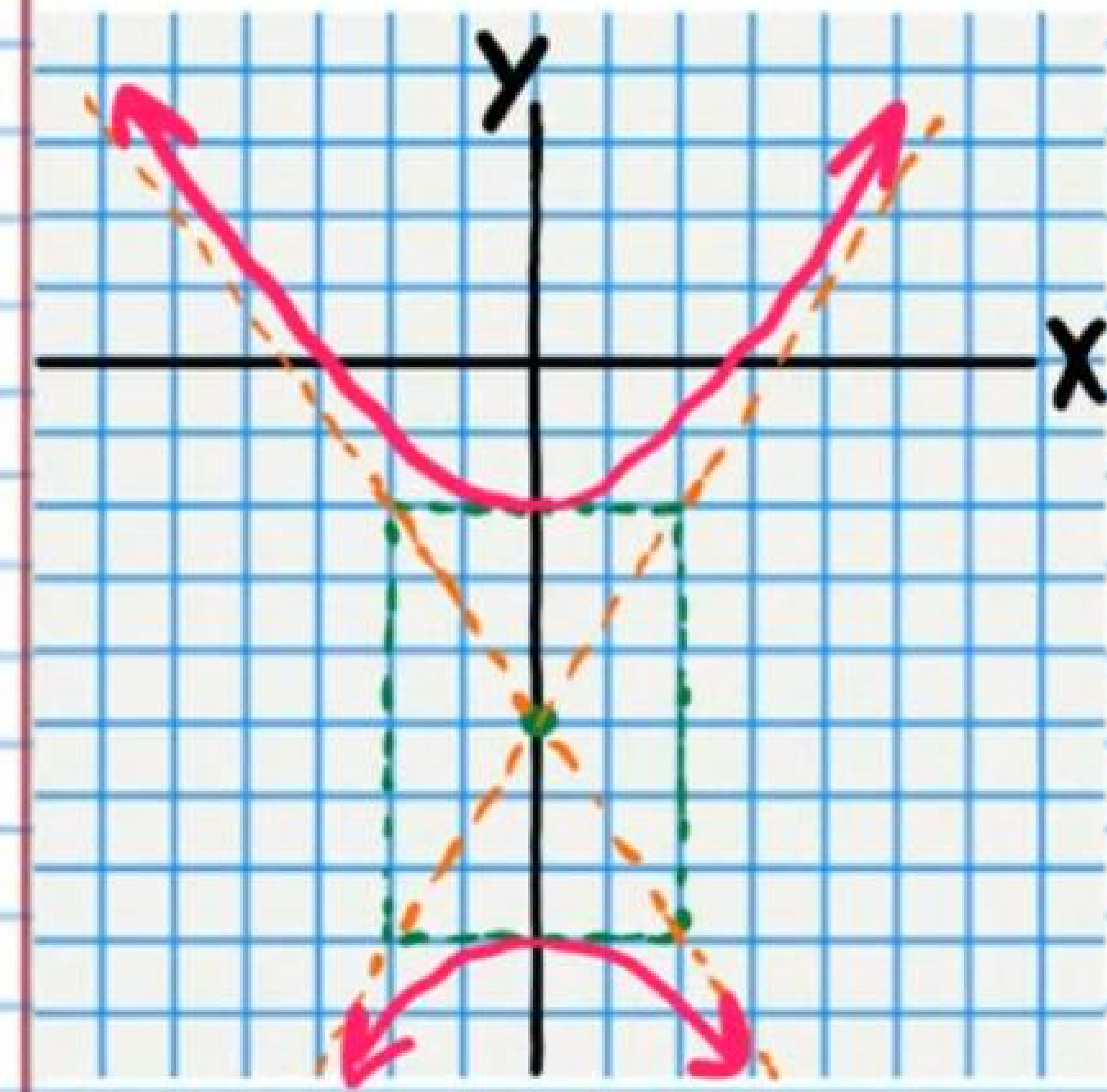


⑬ i) $\boxed{\text{5 people}}$ ii) $\boxed{\text{14 people}}$

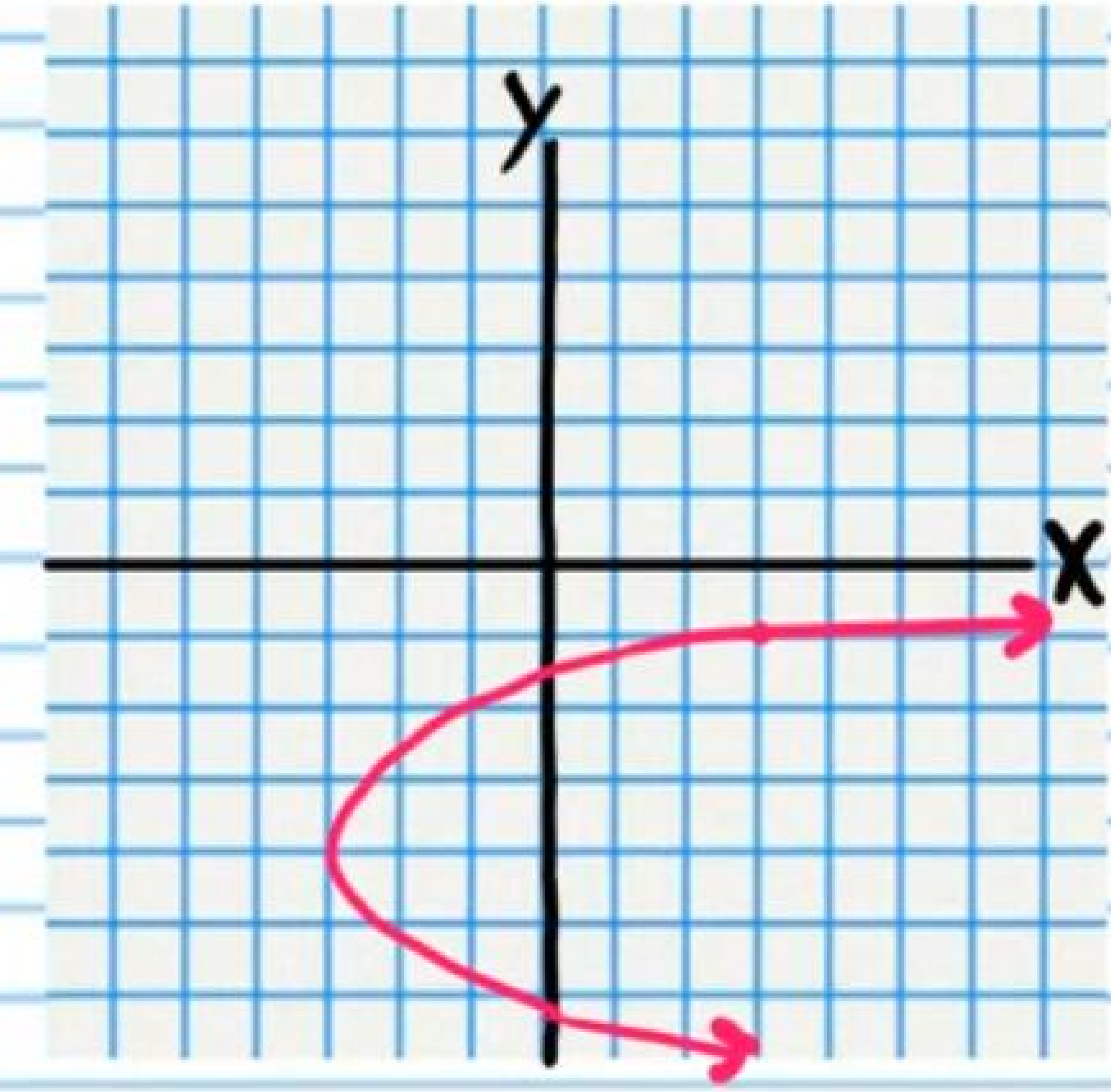
iii) $\boxed{\text{27 people}}$ iv) $\boxed{\text{2 people}}$

v) $\boxed{\frac{22}{29}}$ vi) $\boxed{\frac{2}{29}}$ vii) $\boxed{\frac{8}{29}}$ viii) $\boxed{\frac{27}{29}}$

⑭



⑮



⑯ i) $\boxed{\text{Arithmetic}}$

$\boxed{a_n=19-9n}$ or $\boxed{\{19-9n\}}$

ii) $\boxed{\text{Neither}}$

$\boxed{a_n=n^3}$ or $\boxed{\{n^3\}}$

⑰ i) $\boxed{\text{Arithmetic}}$

$\boxed{a_n=\frac{4}{5}-\frac{1}{5}n}$ or $\boxed{\left\{\frac{4}{5}-\frac{1}{5}n\right\}}$

ii) $\boxed{\text{Geometric}}$

$\boxed{a_n = 0.3\left(\frac{1}{3}\right)^{n-1}}$ or $\boxed{a_n = \frac{3}{10}\left(\frac{1}{3}\right)^{n-1}}$ or $\boxed{a_n = \frac{1}{10}(3)^{2-n}}$

or $\boxed{\left\{0.3\left(\frac{1}{3}\right)^{n-1}\right\}}$ or $\boxed{\left\{\frac{3}{10}\left(\frac{1}{3}\right)^{n-1}\right\}}$ etc.

18 i) $\boxed{S_{39} = -7{,}605}$

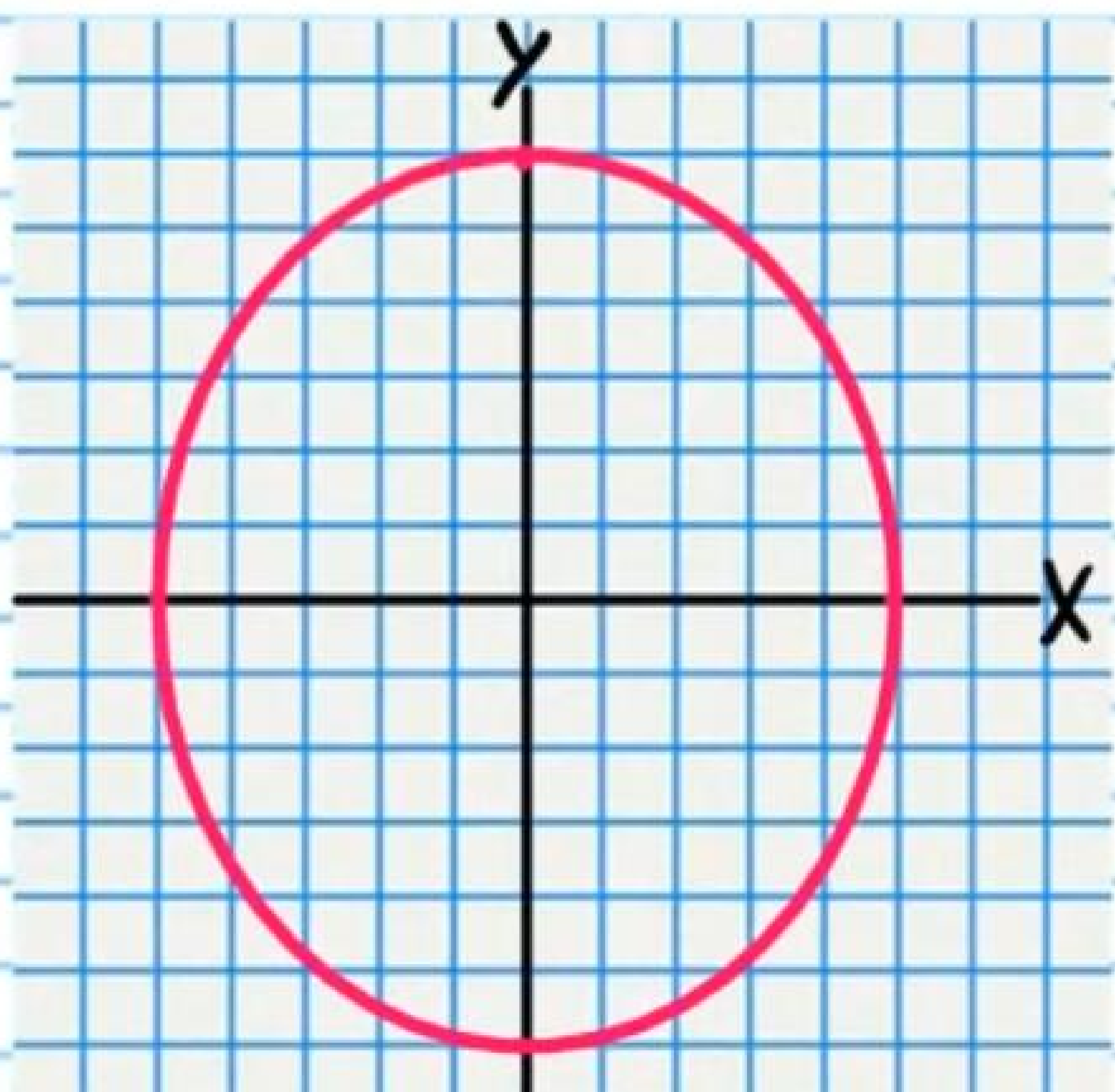
ii) $\boxed{259{,}459{,}200 \text{ outcome variations}}$

19 i) $\boxed{\{-14, -13, -12\}}$ ii) $\boxed{\{-16, -15, -14, -13, -12, -11\}}$

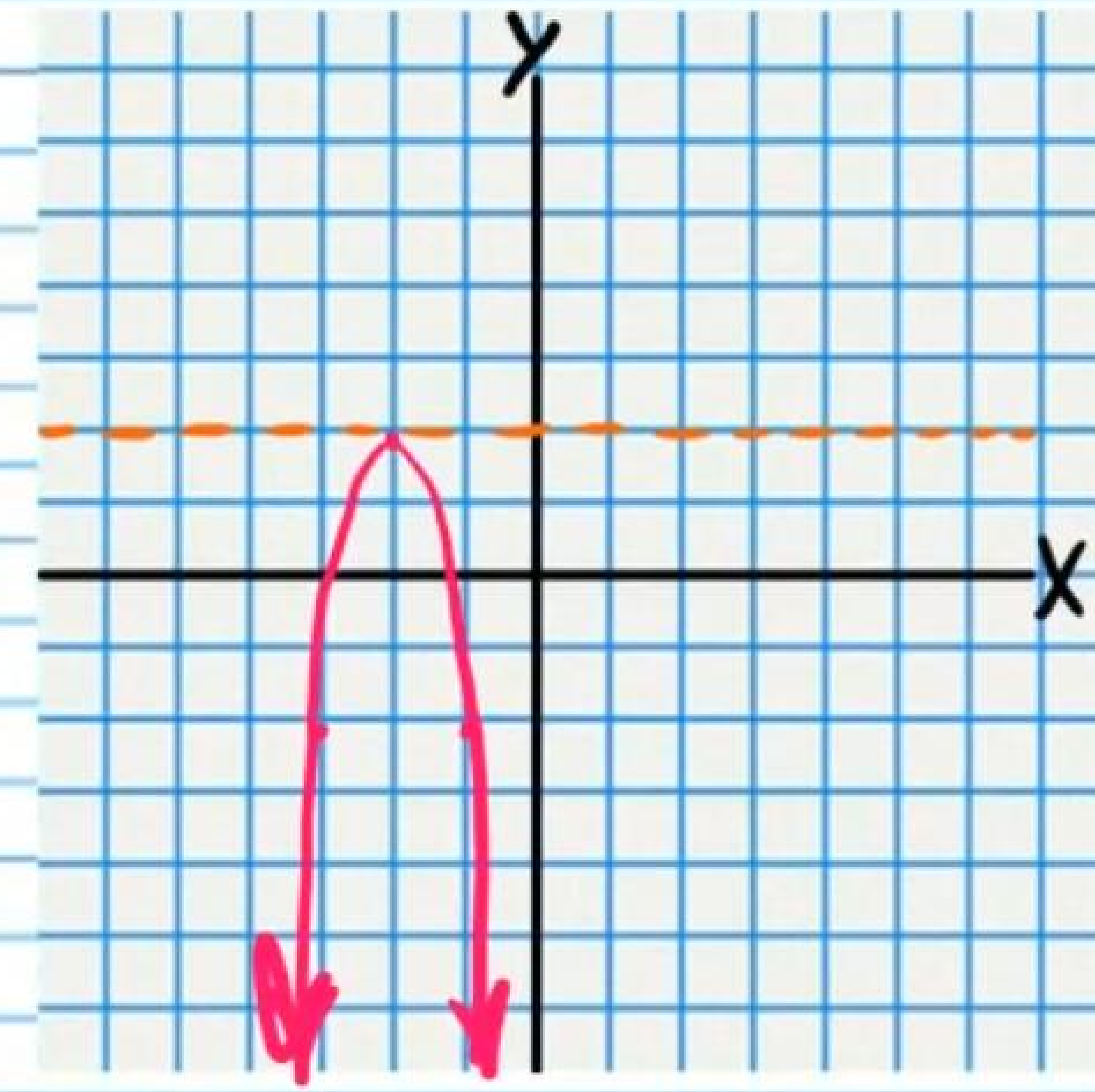
20 $\boxed{\frac{1}{625}}$

21 $\boxed{F_1(0, \sqrt{11})}$

$\boxed{F_2(0, -\sqrt{11})}$



22 $\boxed{-4(x+2)^2 = y - 1\frac{15}{16}}$



23 $\boxed{7 \text{ baskets}}$

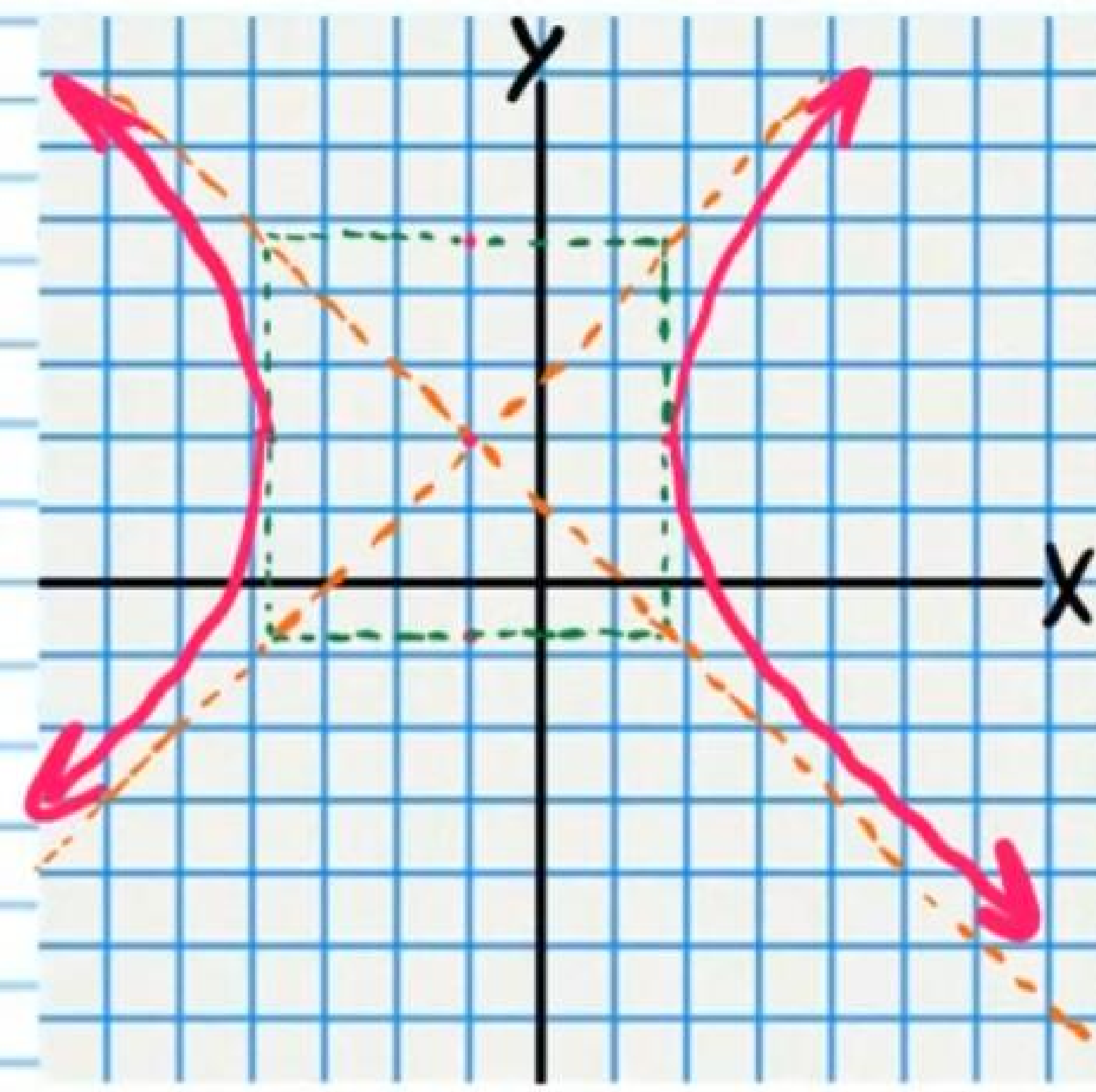
24 $\boxed{F_1(-5, 2)}$

$\boxed{F_2(3, 2)}$

Asymptotes

$\boxed{y - 2 = x + 1}$

$\boxed{y - 2 = -(x + 1)}$



25 i) $\boxed{\frac{5}{3{,}542}}$ ii) $\boxed{\frac{5}{138}}$ iii) $\boxed{\frac{15}{506}}$ iv) $\boxed{\frac{5}{46}}$